Kaffa: Solution Design

An AI compliance team for Ethiopian coffee exporters · Version 2

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Summary

Kaffa gives an Ethiopian coffee exporter a team of AI agents that takes every EU-bound lot from messy farm records to a buyer-ready evidence pack before the 30 December 2026 deadline.

Version 1 automated one person's chasing. Version 2 runs the whole compliance operation across three surfaces:

- **Exporter console:** shipments, risk, approvals and the agents' work as it happens
- **Field app inside Telegram:** field agents walk a farm boundary with GPS, send photos and voice notes in Amharic, and receive a daily visit route
- **Buyer portal:** EU buyers see live readiness, inspect the evidence and download files in the EU's official upload format

Seven specialist agents share one evidence store. Fixed rules decide compliance status, Gemini handles reading, planning and conversation, and a person approves anything with legal or commercial weight. Every action is written to a tamper-evident ledger, because operators must keep due diligence records for five years.

What changed from version 1

Area	Version 1	Version 2
Data intake	CSV upload	Excel files, photos of paper registers, contract PDFs and buyer emails, with duplicate farmers caught across Amharic and Latin spellings
Satellite evidence	One dataset	Several datasets combined into an agreement score, with 2020 and latest-year imagery side by side
Field work	Location pins in chat	GPS boundary walking in a Telegram Mini App, voice notes, and visit routes planned around roads and rain
Supply chain	Plot status only	Chain-of-custody graph and volume checks that catch coffee from unmapped farms

Area	Version 1	Version 2
Shipments	Simple ready-date estimate	Probability of shipping on time, plus a container planner that picks the best compliant lots
Buyers	One approved email	Live buyer portal, files in the EU format, and cited answers to buyer questions
Trust	Approval gates	Autonomy settings per action, tamper-evident ledger, agent traces and an automated evaluation suite

Why this role

The deadline is 54 days away, and the bottleneck is proving compliance, not stopping deforestation.

- **The rule:** the EU deforestation regulation applies from 30 December 2026 for large and medium operators, and 30 June 2027 for micro and small ones. The cut-off date is 31 December 2020.
- **The market:** the EU buys about 30% of Ethiopia's coffee exports, more than any other single market.
- **The gap:** an ODI Global study, as reported by Birr Metrics, finds traceability gaps in up to 57% of supply chains. It estimates up to 60% of smallholders could be excluded.
- **The disagreement:** Ethiopia's coffee authority says over 99% of farms are deforestation-free and disputes the mapping figures as out of date. An ODI consultant says the main challenge is the capacity to demonstrate compliance.

The obligation falls on the EU operator, so the Ethiopian exporter's problem is commercial: a buyer who cannot get clean data for a lot can stop buying it.

What already exists

Screening and paperwork are well served, so Kaffa uses them as inputs instead of competing with them.

Existing tool	What it does	What it leaves open
Botminds	Seven agents for EU compliance teams, including an Outreach agent that chases suppliers in their language	Named languages are French, Portuguese, Bahasa and English. No mention of Amharic, Telegram as an agent channel, or field agents

Existing tool	What it does	What it leaves open
ECTMS	Ethiopia's national system: farm coordinates, supply chain tracking, risk assessment	Coverage and exporter access are not described in the sources read
TechnoServe app	Free Android app that captures farmer data and shares files over WhatsApp or Telegram	File sharing only, with no follow-up workflow
Whisp	Open-source API that analyses plots a user submits	Analysis only, with no compliance decision

Botminds is the closest match and shows a compliance team will pay for agent-led chasing. Kaffa takes the other side of that trade: the exporter in Addis Ababa, working with field agents on Telegram in Amharic. I found no direct equivalent, based on a limited search of public pages.

This role also fits Brain3's premise: an exporter facing this deadline needs a person to do the work more than another dashboard.

Role breakdown

An exporter's compliance work splits into 14 tasks. Kaffa takes on 12, through its agents and its ledger; land legality review and the final EU filing stay with people. Priority reflects how directly a task unblocks a shipment.

Priority	Task	How it is done today	Kaffa agent
1	Turn registers and contracts into records	Retyping paper ledgers and spreadsheets	Intake
1	Validate and repair plot shapes	Checking coordinates by hand	Geo analyst
1	Screen plots for deforestation	Running plots through a tool in batches	Geo analyst
1	Chase field agents for missing data	Phone calls and messages, tracked by memory	Field liaison
1	Plan re-mapping visits	Ad hoc trips	Field planner
1	Forecast and plan shipments	Spreadsheets	Shipment planner

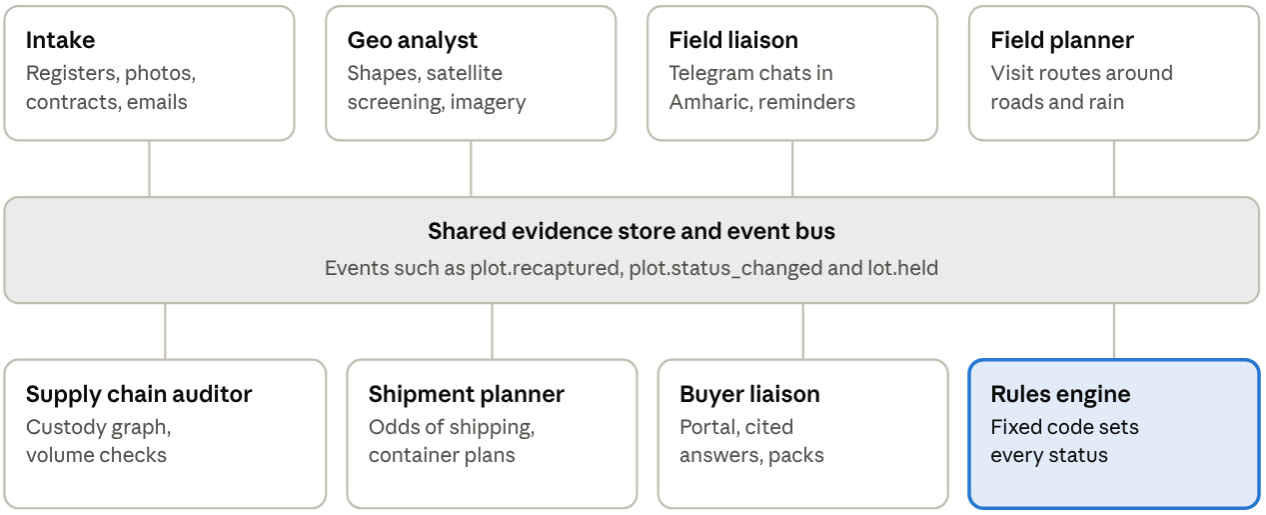
Priority	Task	How it is done today	Kaffa agent
2	Check delivered volumes are plausible	Rarely done	Supply chain auditor
2	Catch duplicate farmers and overlapping plots	Rarely done	Intake and auditor
2	Answer buyer questions	Email back and forth	Buyer liaison
2	Prepare buyer evidence	Assembling files by hand	Buyer liaison
3	Monitor cleared plots for new alerts	Rarely done	Geo analyst
3	Keep records for five years	Shared folders	The platform's ledger
Out of scope	Verify land legality	Reviewing land documents	People, with document storage in Kaffa
Out of scope	File the EU due diligence statement	Done by the EU operator	Kaffa supplies the data in the EU format

Kaffa does not replace the compliance officer. It turns that person into the approver of work the agents have already done.

The AI team

Seven specialist agents work from one shared evidence store. They coordinate through typed events, not free-form chat between models, so every hand-off can be tested and is recorded in the ledger.

Seven agents share one evidence store and coordinate through events



The highlighted rules engine is code, not an agent. It is the only part that can change a plot or lot status.

the AI team · seven agents, one rules engine

Agent	Job	Acts alone on	Asks a person before
Intake	Reads registers, photos, PDFs, contracts and buyer emails into records, and resolves duplicate farmers	Draft records, with a confidence score for every field	Merging two farmers below the match threshold
Geo analyst	Validates and repairs plot shapes, screens them against several datasets, and monitors cleared plots	Checks, and repairs that do not move a boundary, such as closing a ring	Any repair that changes a plot's shape
Field liaison	Holds Telegram conversations in Amharic, collects boundaries, photos and voice notes, then reminds and escalates	Messages to registered field agents, within a daily cap	Messaging anyone new
Field planner	Groups plots that need visits and plans routes around roads and rain	Drafting day plans	Sending a plan, unless set to automatic

Agent	Job	Acts alone on	Asks a person before
Supply chain auditor	Builds the custody graph, checks volumes and finds anomalies	Raising flags, which hold lots automatically	Clearing a flag
Shipment planner	Forecasts readiness, allocates lots to containers and proposes substitutes	Forecasts and draft plans	Changing an allocation
Buyer liaison	Runs the buyer portal, answers questions with citations, triages buyer email and prepares evidence packs	Answers built only from stored evidence, if the exporter allows it	Sending any email or evidence pack

A coordinator routes events between the agents. A re-walked boundary, for example, sets off this chain:

1. Field liaison records `plot.recaptured`.
2. Geo analyst re-validates and re-screens the plot, then records `plot.status_changed`.
3. Supply chain auditor re-runs the volume check for every lot that uses the plot.
4. Shipment planner updates the forecast and the container plan.
5. Buyer liaison refreshes the buyer portal.

Each agent is a system prompt, a typed tool set and its own evaluation set, so one agent can change and be tested without touching the others.

Key workflows

Six workflows cover the job end to end. Each runs as a durable workflow, so a step that waits two days for a reply survives restarts and deployments.

From paper register to verified plot

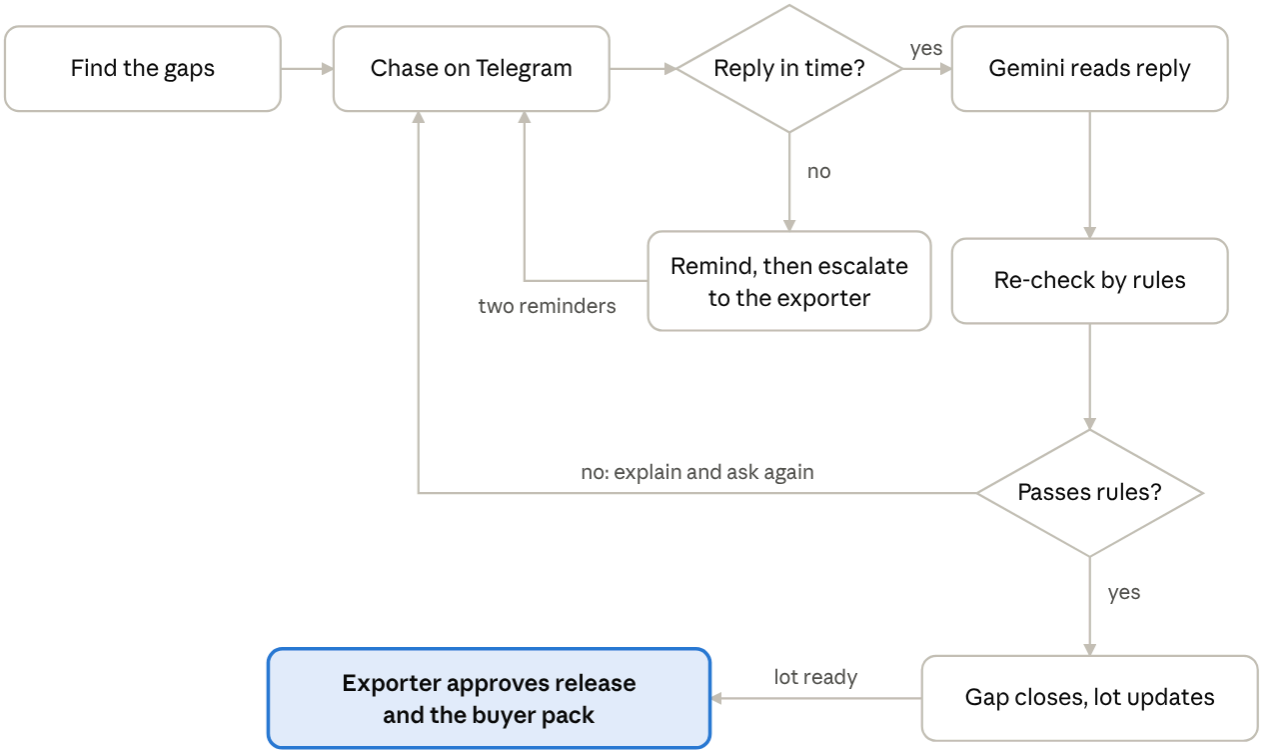
1. The exporter uploads an Excel file or a phone photo of a register page.
2. The intake agent extracts farmers, plots and sizes, with a confidence score for each field. Low-confidence fields go to a review queue.
3. Identity resolution matches each name against existing farmers across Amharic and Latin spellings, plus phone numbers and village. It proposes merges instead of creating duplicates.

4. The geo analyst checks each plot against the EU's GeoJSON rules, repairs what it safely can, and screens valid plots.
5. Plots that need field work go into the chase loop.

The chase loop

The field liaison sends one Telegram message per field agent, in Amharic, listing exactly which farms need what. The workflow then waits for the reply event. With no reply after 48 hours it sends a reminder, and after 96 hours it escalates to the exporter. A reply can be a boundary walked in the Mini App, a location pin, a photo, text or a voice note.

A gap closes only after the rules pass the reply



chase loop · from gap to buyer pack

The highlighted step is the only one that releases a lot or sends anything to a buyer.

Field visit planning

Plots that need a physical re-walk are grouped by area. The field planner gets road travel times from openrouteservice and orders the stops with its own route solver. It moves visits away from forecast heavy rain using Open-Meteo, then sends each field agent a day plan in Telegram, with an optional Google Calendar event.

Shipment planning

A contract PDF or buyer email becomes a shipment with volume, grade, port and ship date. The shipment planner allocates verified lots to containers and forecasts the chance of shipping on time. When a lot is held, it proposes substitute lots and shows how each changes that chance.

Buyer collaboration

Each buyer gets a portal link. They see readiness per lot, inspect the evidence, download files in the EU format and ask questions. The buyer liaison answers from stored evidence and the regulation, with citations, and sends anything uncertain to the exporter. Buyer emails in the exporter's inbox become tasks automatically.

The daily run

Each morning a scheduled job re-screens cleared plots for new alerts, reopens affected lots, sends due reminders and re-runs every forecast. The exporter then gets a brief in Telegram and the console with the three decisions that matter most that day.

Signature capabilities

Nine capabilities carry most of the product's value. Each is built so a reviewer can test it, not just watch it in a demo.

1. Evidence from several sources, with an agreement score

Each plot is screened against Global Forest Watch tree cover loss, Global Forest Watch's integrated deforestation alerts and, where a key is available, Whisp, the open-source screening API from FAO and partners. Each source gives its own verdict, and Kaffa reports how far they agree. Disagreement creates a review item rather than a guess.

The reviewer sees the plot on Sentinel-2 cloudless imagery from 2020 and the latest year, with a slider between them. A stretch goal adds vegetation-index change computed from raw Sentinel-2 scenes in the public Earth Search catalogue.

2. Walking the boundary inside Telegram

A Telegram Mini App uses the location access Telegram added in Bot API 8.0. The field agent walks the farm edge and the app:

- Drops GPS points with poor accuracy
- Simplifies the track and shows the live area
- Closes the shape and checks it against EU rules before upload

- Attaches geotagged photos
- Queues everything offline until signal returns

3. Exports in the EU's official format, with repair

Exports follow the EU's GeoJSON specification:

- Country codes and production place names in the expected properties
- Coordinates checked after the EU system's 6-decimal truncation, so a shape cannot turn invalid on upload
- Holes removed by splitting a ring into two halves, the method the EU documentation suggests
- Files split to stay under the 25 MB limit

Kaffa also produces a due diligence data sheet shaped to the EU operator API, so the buyer can file without retyping anything.

4. Identity resolution across scripts

The same farmer may appear as ተስፋዬ in one register and Tesfaye or Tesfaie in another. Kaffa transliterates Ge'ez script to Latin, normalises spelling variants, and scores matches on name, village and phone number. Each merge it proposes comes with its reasons. This stops double-counted farmers from inflating the volume a lot appears to support.

5. Volume checks and chain of custody

Kaffa builds a custody graph from farm to collector, washing station, lot and container. For each lot it compares delivered volume with what the source plots could plausibly produce: area times a regional yield. The national average is about 0.9 tonnes per hectare, from USDA figures reported by Qahwa World. A lot that exceeds its plausible output is flagged as possibly containing coffee from unmapped farms.

6. Probability of shipping on time

Instead of a single date, the shipment planner runs 10,000 simulations. Each draws from the observed response and fix times of the field agents involved. The result is a chance of shipping on time, plus median and worst-case ready dates. The exporter can test changes, such as adding a field visit on Tuesday, and see the probability move.

7. Container planner

The planner treats allocation as an optimisation problem. It picks lots that fill each contract's volume and grade by the ship date, favouring lots with stronger evidence and higher value. When a lot is held, it re-solves and proposes the smallest change that keeps

the shipment whole.

8. Tamper-evident ledger

Every event is stored with a SHA-256 hash that includes the previous event's hash, so editing history breaks the chain. Each evidence pack carries the latest hash, and a public page re-computes it on demand. Each lot also gets a QR code that roasters can scan to verify its claims.

9. Regulation copilot

The regulation, its amendment and Commission guidance are split into passages, embedded with Gemini and stored in Postgres. Every agent answer about the rules must cite a passage. If no passage supports a claim, the agent says it does not know and routes the question to a person.

Trust and human control

Gemini never decides whether a plot or lot is compliant. Fixed code sets every status, and each result records the rule version and dataset version behind it.

Plot status rules

The regulation sets no tolerance, so any detected loss needs a person's attention:

Status	Meaning	Effect on the lot
Clear	Valid shape, sources agree there is no loss after 2020 and no recent alert	Can ship
Review	Loss or an alert below the blocked threshold, or sources disagree	Held until a person accepts it, with a reason
Blocked	Loss covering 10% of the plot or 0.5 hectares, whichever is lower	Held until the plot is removed or disproved
Not checked	Invalid shape, or satellite data unavailable	Held

The two thresholds are design choices, not legal figures. They live in versioned configuration.

Autonomy settings

The exporter sets each action type to automatic, ask first or off. Actions with legal weight are locked to ask first.

Action	Default	Can be made automatic
Chase or remind a registered field agent	Automatic	Already automatic
Send a day plan to a field agent	Automatic	Already automatic
Merge two farmer records	Ask first	Yes, above a set match score
Answer a buyer question from stored evidence	Ask first	Yes
Change a container allocation	Ask first	Yes
Clear a review plot	Ask first	No
Release a lot	Ask first	No
Send an evidence pack or email to a buyer	Ask first	No

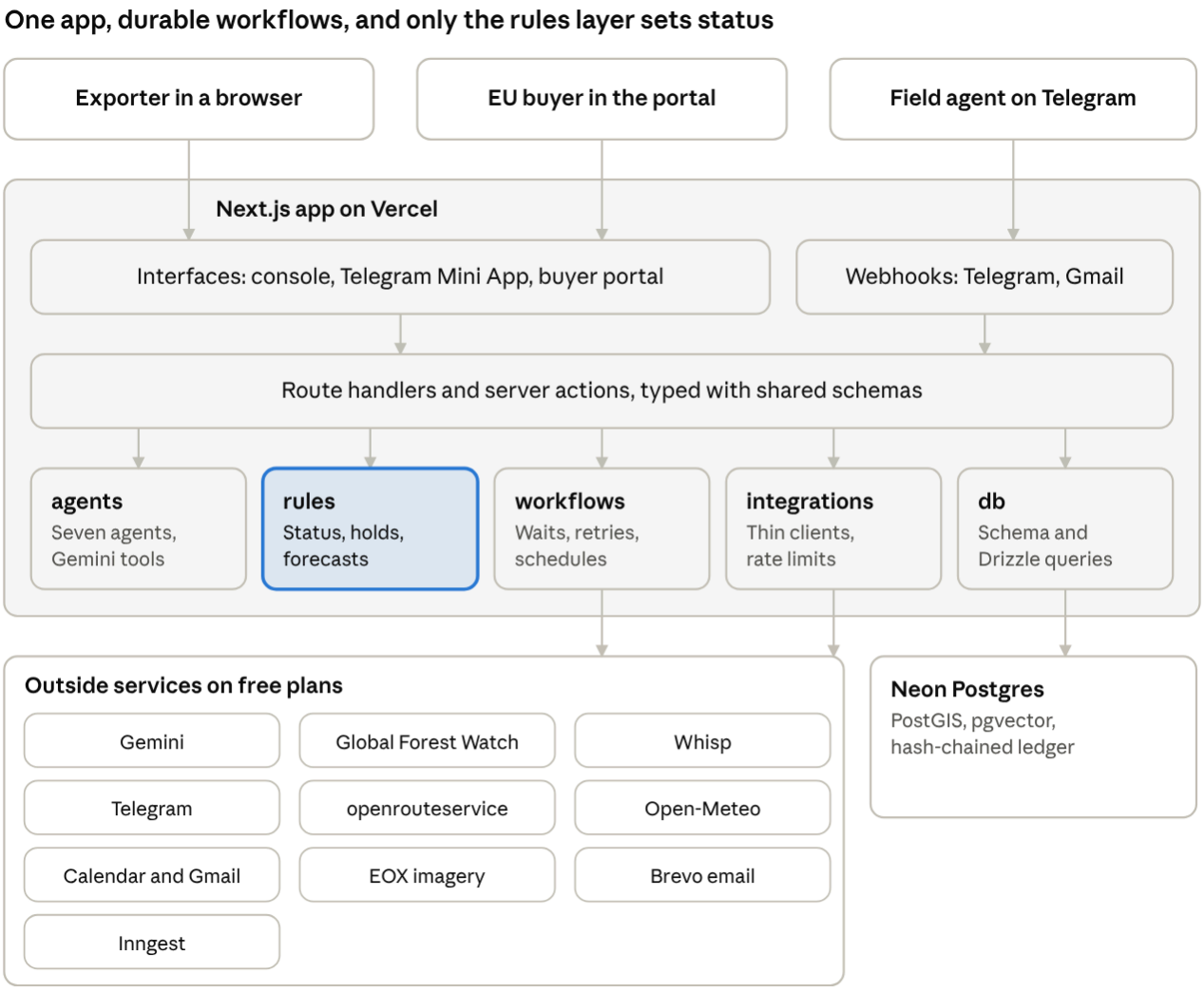
Safeguards

- **Untrusted inputs:** messages, photos, documents and emails are data. Gemini only extracts fields from them, its output goes to the rules, and the evaluation suite includes injection attempts.
- **Known senders:** field updates are accepted only from registered Telegram accounts, and only for that person's cooperative.
- **Tenant isolation:** every query is scoped to one exporter, with separate roles for admin, compliance officer, field agent and buyer.
- **Personal data:** Google's terms say not to send personal information to the unpaid Gemini tier, so the demo uses synthetic farms. Production needs the paid tier.

Kaffa's output supports the buyer's due diligence. It is not a legal determination of compliance.

Architecture

Kaffa stays one Next.js application on Vercel. Durable workflows handle anything that waits, and Postgres holds state, map shapes, search vectors and the ledger. One codebase serves all three surfaces, so the console, the Mini App and the buyer portal share types and rules.



Kaffa architecture · one app, four code layers

The highlighted rules layer is the only code that can change a status. Agents read its results but cannot write them.

Frontend

- **Console and buyer portal:** Next.js with React, TypeScript and Tailwind. Live updates arrive as server-sent events from the event log.
- **Maps:** MapLibre, with the 2020 and latest-year imagery layers and a comparison slider.
- **Custody graph:** React Flow.
- **Telegram Mini App:** a separate route in the same app, with offline storage for boundary walks.
- **Languages:** English and Amharic for the field app.

Backend

- **API:** route handlers and server actions, with Zod schemas shared between client and server.
- **Durable workflows:** Inngest. A chase waits for a reply event with a timeout, so a two-day wait uses no compute and survives deployments. The free plan includes 50,000 executions a month and runs of up to 30 days, with no card.
- **Agents:** Gemini function calling with structured output. Each tool validates its input on the server.
- **Optimisation:** a linear-programming solver for container plans, plus a nearest-neighbour route solver refined with 2-opt swaps.
- **Simulation:** seeded Monte Carlo runs, so every forecast can be reproduced.

Data

Neon Postgres supports both PostGIS and pgvector. PostGIS finds overlapping plots in the database, and pgvector stores the regulation passages.

Area	Main tables
Organisations	`orgs`, `users`, `roles`, `autonomy_policies`
Supply	`cooperatives`, `field_agents`, `farmers`, `farmer_aliases`, `plots`, `plot_versions`
Evidence	`checks` with source, dataset version and rule version, `media`, `documents`
Trade	`contracts`, `shipments`, `containers`, `lots`, `deliveries`, `custody_edges`
Work	`gaps`, `tasks`, `conversations`, `messages`, `approvals`
Trust	`ledger_events` with hash and previous hash, `agent_runs`, `agent_steps`
Knowledge	`reg_passages` with embeddings

Observability and quality

Every agent run is saved as a trace of its steps, tool calls, inputs, outputs, timing and token use, and the console shows it. An evaluation suite runs in continuous integration on every change. It is described under testing below.

Third-party APIs and free-tier limits

Every service runs on a free plan with no card. Each one sits behind a thin client with its own rate limiter, so hitting a limit slows Kaffa down instead of breaking it.

Service	Used for	Free-tier position	To confirm during build
Gemini	Reasoning, extraction from photos and PDFs, Amharic messages, voice notes, embeddings	Free tier with limits shown in AI Studio. Ethiopia is a supported region	Best free model with function calling. Afaan Oromo support for voice
Global Forest Watch	Tree cover loss and integrated deforestation alerts	Free API key with polygon queries	Current dataset versions
Whisp	Second opinion built on several forest datasets	Hosted API with a personal key	Whether the key is free. Kaffa works without it
Telegram Bot API and Mini Apps	Chat, boundary walking, location	No charge for messaging users. Location access in Mini Apps	None
openrouteservice	Road travel times between visits	Free key with daily quotas	Exact matrix quota
Open-Meteo	Rain forecast for visit planning	Free for non-commercial use, under 10,000 calls a day	None
EOX Sentinel-2 cloudless	2020 and latest-year imagery	Free for non-commercial use with attribution	A commercial launch needs a licence or another source
Google Calendar and Gmail APIs	Visit events, buyer email intake	No charge. Gmail read access is a restricted scope	Access in testing mode for demo accounts
Inngest	Durable workflows	50,000 executions a month, no card	None

Service	Used for	Free-tier position	To confirm during build
Brevo	Outbound email	Transactional send API	Free allowance. Gmail is the fallback
Neon	Postgres with PostGIS and pgvector	1 GB storage, 100 compute hours, no card	None
Vercel	Hosting and the daily job	300-second functions, daily cron	None

Scheduler decision

The daily run uses Vercel Cron, and multi-day waits use Inngest. Both call secured HTTP endpoints, so Google Cloud Scheduler can replace either with no code change.

User interface

Each of the three users gets a surface built for where they work: the exporter at a desk, the field agent on a phone in the field, and the buyer in Europe.

Exporter console

Screen	What it shows	What the user can do
Command centre	Days to the deadline, shipments ranked by risk, chance of shipping on time, the morning brief	Open a shipment, run the daily job
Shipment and containers	Lots by container, held lots, proposed substitutes, forecast	Accept a plan, test what-if changes
Map and evidence	Plots coloured by status, 2020 and latest-year imagery with a slider, each source's verdict	Inspect a plot, accept a review with a reason
Supply chain	Custody graph from farm to container, volume checks	Trace a lot, clear a flag
Field operations	Each field agent's conversations with English beside the Amharic, open items, day routes	Take over a conversation, adjust a route

Screen	What it shows	What the user can do
Intake review	Extracted fields below the confidence threshold, proposed farmer merges	Correct a field, approve or reject a merge
Approvals	Everything waiting for a person, with what will happen if approved	Approve, reject or edit
Agent activity	Every agent run as a trace of steps and tool calls	Filter by agent, shipment or field agent
Settings	Autonomy level per action, thresholds, team and roles	Change policies, invite people

Ask Kaffa sits on every screen. It answers questions such as "Why is the Hamburg shipment at risk?" with links to the evidence, and takes instructions within the autonomy settings.

Field app in Telegram

The field agent never installs anything. The chat carries reminders and replies in Amharic, including voice notes, and the Mini App opens from a button for four tasks:

- See today's visits in route order
- Walk a boundary
- Take geotagged photos
- Review open items

Buyer portal

Buyers sign in through a magic link. They see readiness per lot, open the evidence for any plot, and download files in the EU format. They can also ask questions and check any evidence pack against the ledger. A public page behind each lot's QR code shows a read-only summary.

Business case

The exporter pays, because a lot without proof is a lot an EU buyer cannot place on the market, and the EU takes about 30% of Ethiopia's coffee exports.

Why hire Kaffa over a person

- It works with hundreds of field agents at once, in their language, and never misses a follow-up.
- It screens every plot overnight and re-checks cleared plots every day.
- It catches what people rarely check: duplicate farmers, impossible volumes and overlapping plots.
- It writes the audit trail as it works, instead of rebuilding it when an authority asks.

The compliance officer stays, but spends their time on decisions instead of chasing.

Pricing hypothesis

- **Exporters:** a seasonal subscription priced by verified tonnes, so cost tracks the value at stake
- **Cooperatives and field agents:** free, because their participation is what makes the data possible
- **Buyers:** free, because each buyer can invite its other origin suppliers

That last point is the growth loop. One EU roaster buys from many exporters, and each invitation brings in another paying exporter. Price points would come from interviews with exporters, so none are set here.

Success metrics

Metric	What it shows
Share of EU-bound volume buyer-ready by its ship date	The core outcome
Median days from gap found to gap closed	How well the chase works
Field agent replies within 48 hours	Whether Telegram in Amharic fits the field
Review flags people overturn	False-alarm rate of screening
Intake fields people correct	Extraction accuracy
Buyer questions answered without the exporter	Value on the buyer side

Failure handling, testing and evaluation

Kaffa fails closed: when something breaks, a plot stays not checked and a lot stays held. It never shows a pass it did not earn.

Failure	Behaviour
A satellite source is down or rate limited	Other sources still report, the agreement score shows the gap, and the plot cannot be cleared on fewer sources than its policy requires
Gemini is over quota	Requests queue behind a rate limiter. Checks and holds continue, and chases fall back to fixed Amharic and English templates
A workflow step fails	Inngest retries with backoff, then raises a task for a person
Duplicate Telegram or email events	Ignored, using each message's ID
GPS accuracy is poor during a walk	Weak points are dropped. If too few remain, the app asks for a re-walk
A reply cannot be understood	One clarifying question, then escalation after a second failure
Road travel times are unavailable	Straight-line distances, marked as estimates on the route
The daily job is missed	The next run catches up, and the console shows the last successful run

Testing

- **Rules:** unit tests written before the code, including the EU edge cases: holes, figure-eight shapes, shapes that collapse after 6-decimal truncation, and the 4-hectare point rule
- **Algorithms:** seeded tests for the simulation, the container solver, the route solver and transliteration matching
- **Integrations:** recorded responses for every outside service, plus webhook tests for Telegram and Gmail
- **End to end:** browser tests of the full path from a register photo to a buyer pack

Agent evaluations

Each agent has a golden set that runs in continuous integration, and a drop in score blocks the change:

Agent	What is scored
Intake	Field accuracy on synthetic register photos and spreadsheets
Intake	Precision and recall of farmer matches across scripts
Field liaison	Correct extraction from Amharic replies, plus resistance to injected instructions
Buyer liaison	Every claim carries a citation, and unsupported questions are passed to a person
Shipment planner	Plans meet volume, grade and date constraints

Demo honesty

- Farms, cooperatives and buyers are synthetic and labelled as such.
- Satellite results are real, because the synthetic plots sit at real locations.
- New forest alerts cannot appear on cue, so the daily-run demo uses a labelled test alert.
- A native speaker checks the Amharic templates before the demo.

Scope and build order

The build runs in three tiers, so a complete product exists at the end of each one.

Tier 1: the core loop

1. Data model, rules, ledger and their tests
2. Intake from Excel and register photos
3. Global Forest Watch screening, imagery slider and EU-format export
4. Telegram bot, the boundary-walking Mini App and the durable chase workflow
5. Console: command centre, map and evidence, approvals, agent activity
6. On-time probability forecast
7. Read-only buyer portal with downloads

Tier 2: the depth

1. Identity resolution across scripts
2. Custody graph and volume checks
3. Container planner
4. Field visit planner with routes, rain and calendar
5. Regulation copilot and cited buyer answers
6. Gmail intake
7. Evaluation suite in continuous integration

Tier 3: if time allows

- Whisp as an extra evidence source
- Vegetation-index change from raw Sentinel-2 scenes
- Voice notes and Afaan Oromo
- Public QR verification pages

Done means: a reviewer watches a photo of a paper register become verified plots, a field agent fix a failing boundary by walking it in Telegram, the on-time probability rise, and an approved evidence pack reach the buyer portal.

Not in this version

- Land legality review, beyond storing documents
- Filing statements to the EU system, which needs the EU operator's credentials
- Integration with Ethiopia's national system, which has no public API in the sources read
- Billing